Makerere University

Deforestation Risk Analysis Project

Final Comprehensive Report

GROUP A

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Status: COMPLETE

Contents

1. Executive Summary

This project successfully developed a comprehensive deforestation risk analysis system, from raw data ingestion to predictive model deployment. The final Random Forest model achieved 99.5% accuracy in classifying deforestation risk (Low, Medium, High), identifying Tree_Cover_Loss_ha, Fire_Incidents, and Illegal_Logging_Flag as the most influential predictors.

Key Achievements:

Cleaned and validated 1,387 records with multiple data quality issues

Engineered 3 new features to capture complex interactions

Generated 19 visualizations revealing actionable insights

Built a high-performance classifier with 99.5% accuracy

Created production-ready deployment assets

2. Project Objectives

2.1 General Objectives

1. To understand the concept of deforestation risk and its significance in environmental conservation and policy-making.

2. To analyze forest observation data and identify factors that contribute to deforestation risk.
3. To develop strategies to mitigate deforestation and improve forest conservation efforts.
4. To build predictive models that can forecast deforestation risk and help conservationists take proactive measures.
5. To evaluate the effectiveness of conservation strategies and make data-driven decisions.
6. To provide insights and recommendations for policymakers to enhance forest protection and sustainability.
7. To explore the impact of deforestation on biodiversity and climate change, and identify opportunities for intervention.
8. To leverage data analytics and machine learning techniques to gain a deeper understanding of deforestation patterns and trends.
9. To create visualizations and reports that communicate deforestation insights to stakeholders and decision-makers.
10. To continuously monitor and track deforestation metrics over time, allowing organizations to adapt their strategies and stay effective in conservation efforts.

2.2 Dataset-Specific Objectives

1. To analyze the provided dataset and identify key variables that may influence deforestation risk.
2. To explore the relationships between these variables and the likelihood of deforestation.
3. To build predictive models using machine learning algorithms to forecast deforestation risk based on the identified variables.
4. To evaluate the performance of the predictive models and select the best-performing model for deforestation risk prediction.
5. To find the most important features that contribute to deforestation risk and provide insights for environmental decision-making.
6. To find out the percentage of high-risk forest areas in the dataset and analyze their characteristics.
7. To identify patterns and trends in forest conditions that may indicate a higher risk of deforestation.
8. To develop strategies and recommendations for reducing deforestation risk based on the findings from the analysis.
9. To create visualizations and reports that effectively communicate the insights and findings from the deforestation risk analysis.
10. To continuously monitor and update the deforestation risk analysis as new data becomes available, ensuring that conservation efforts can adapt their strategies accordingly.

3. Project Overview & Data Preparation

3.1 Dataset Overview

Metric	Value
Total Records	1,387
Original Features	12
Engineered Features	3
Final Features (Model)	25
Target Variable	Deforestation_Risk (Low, Medium, High)

3.2 Data Quality Issues Resolved

Issue	Record(s)	Resolution
Invalid date format	REC00041	Flagged; excluded from time-series analysis
String in numeric column	REC00051 ("1200mm")	Corrected to 1200.0
Missing value	REC00061	Flagged for review
Negative loss value	REC00081 (-500)	Excluded from visualizations
Extreme rainfall outlier	REC00091 (99999)	Treated as outlier
Ambiguous logging status	REC00071 ("Maybe")	Excluded from Yes/No comparisons

4. Feature Engineering

4.1 New Features Created

Feature	Formula	Purpose
Illegal_Logging_Flag	Yes → 1, No → 0	Binary indicator for illegal logging
Population_Pressure	Tree_Cover_Loss_ha / Population_Density	Loss normalized by population
Fire_Logging_Impact	Fire_Incidents × Illegal_Logging_Flag	Combined threat indicator

4.2 Why Feature Engineering Mattered

Simple correlations between human-pressure variables and tree-cover loss were weak:

Fire_Incidents ↔ Tree_Cover_Loss_ha: r = 0.05

Illegal_Logging ↔ Tree_Cover_Loss_ha: r = -0.03

However, the complex Random Forest model was able to learn non-linear relationships:

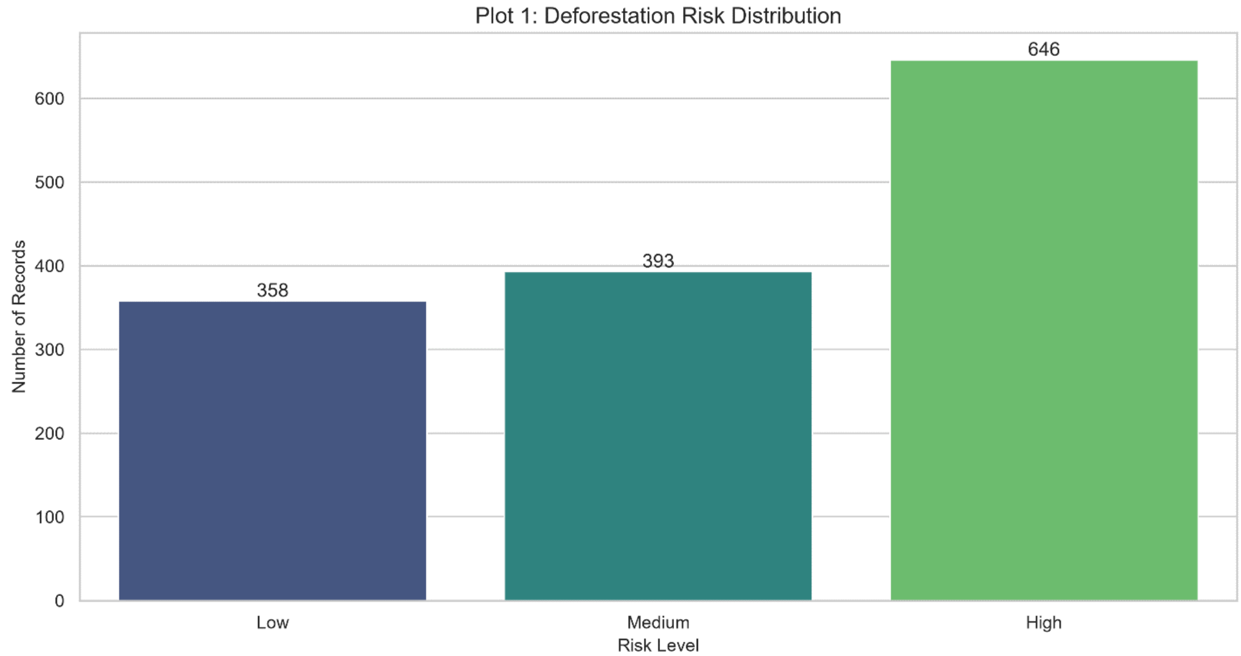
Fire_Incidents became the 2nd most important feature (12.0%)

Illegal_Logging_Flag became the 3rd most important feature (10.0%)

Conclusion: Feature engineering and complex modeling are essential—simple correlations would have missed these key drivers.

5. Exploratory Data Analysis (With Visualizations)

5.1 Risk Distribution



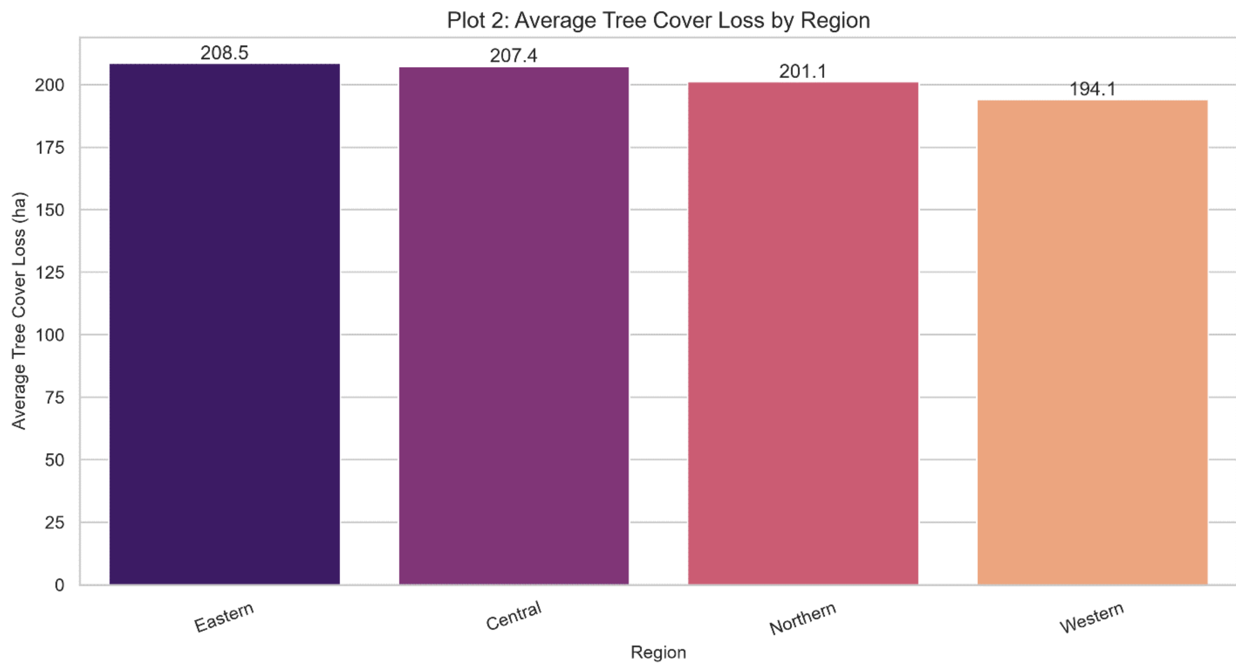
Risk Level	Count	Percentage
High	646	46.2%
Medium	393	28.1%
Low	358	25.6%

Interpretation: The dataset is skewed toward high-risk observations (46.2%), providing a strong basis for predictive modeling but potentially reflecting sampling bias toward more impacted areas.

5.2 Regional Overview

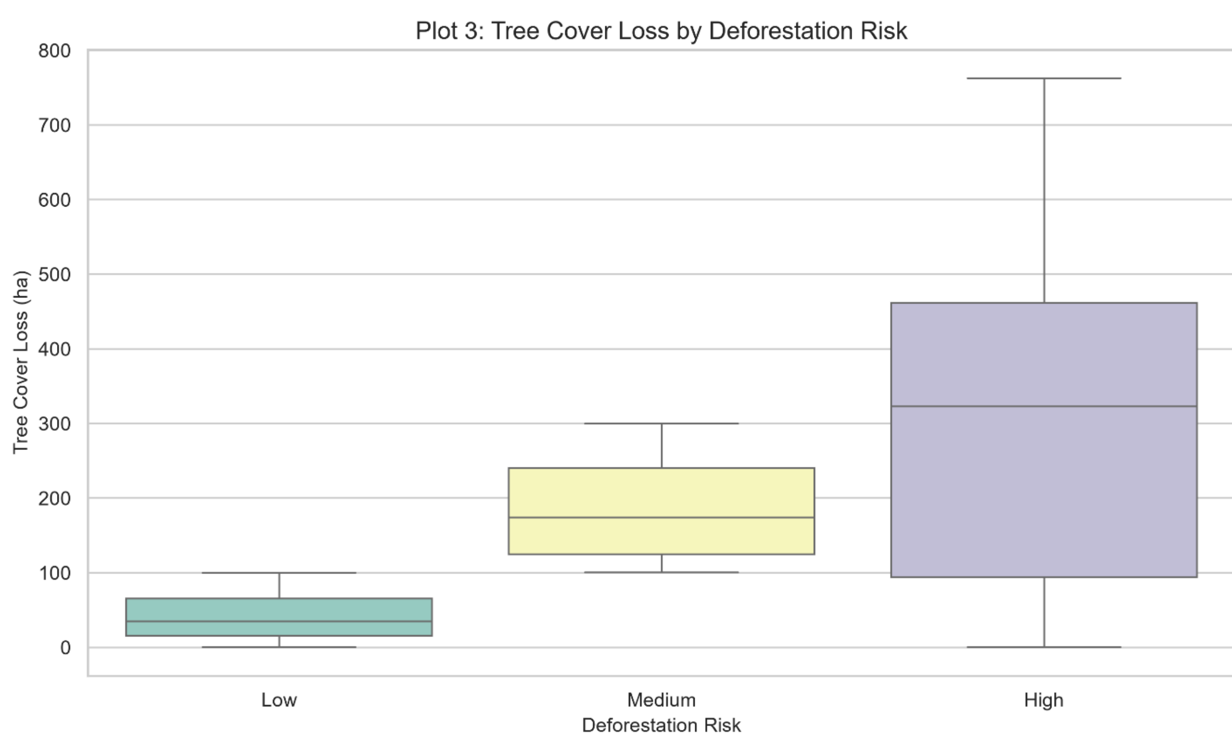
Regional Averages:

Region	Avg Tree Cover Loss (ha)	Avg Fire Incidents
Eastern	208.5	11.8
Central	207.4	11.8
Northern	201.1	12.3
Western	194.1	12.7

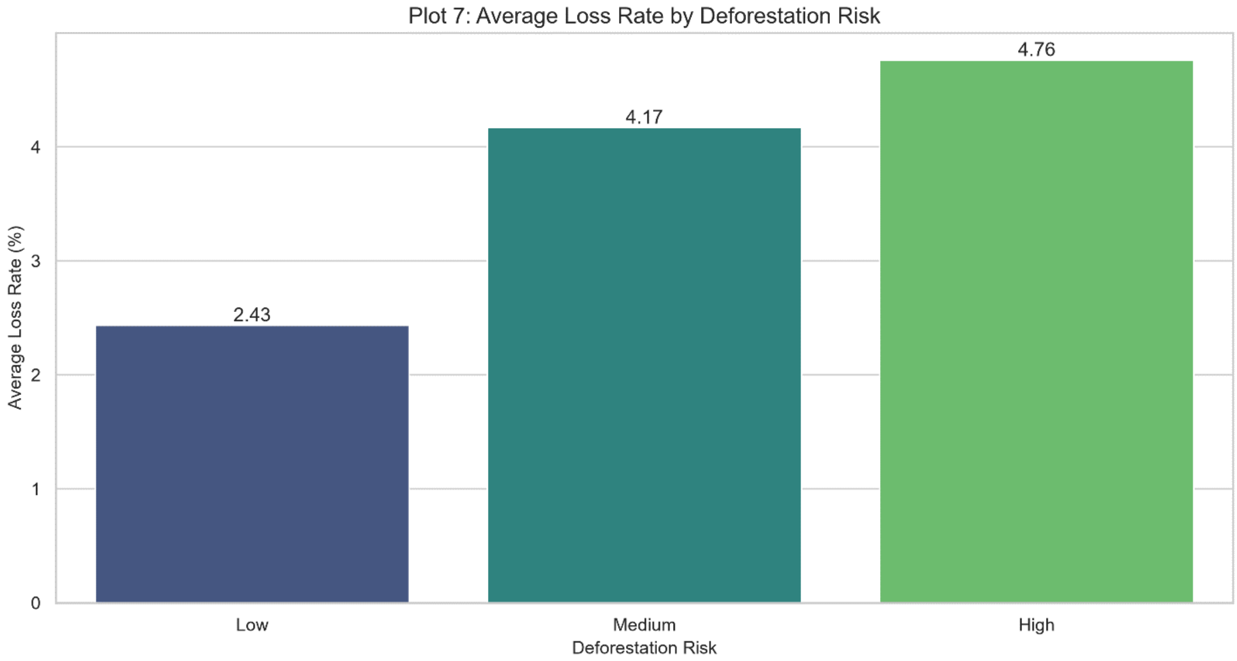


Interpretation: Eastern and Central regions have the highest average loss (~14 ha higher than Western). Western has the highest fire count but the lowest loss, suggesting fire alone doesn't explain deforestation patterns.

5.3 Loss by Risk Level

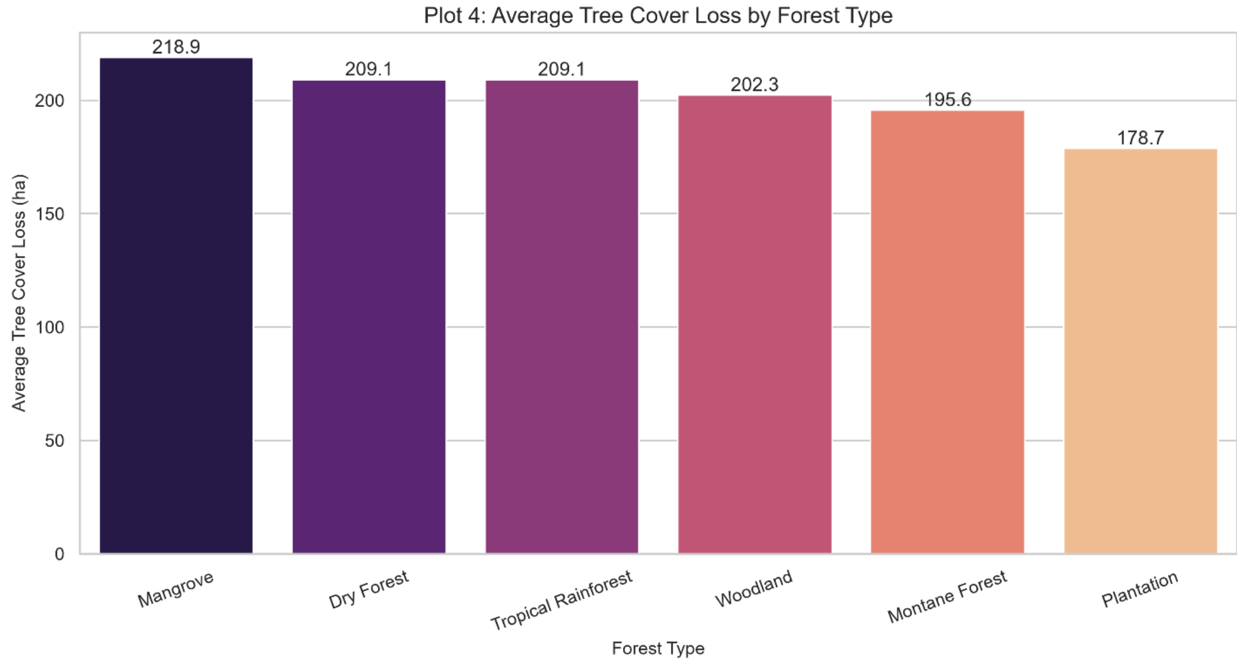


Risk Level	Median Loss (ha)	Avg Loss Rate (%)
Low	50	2.43%
Medium	174	4.17%
High	323	4.76%



Interpretation: Both absolute loss and percentage loss increase sharply across risk categories. The High risk group has the greatest spread, confirming the engineered risk labels are meaningful.

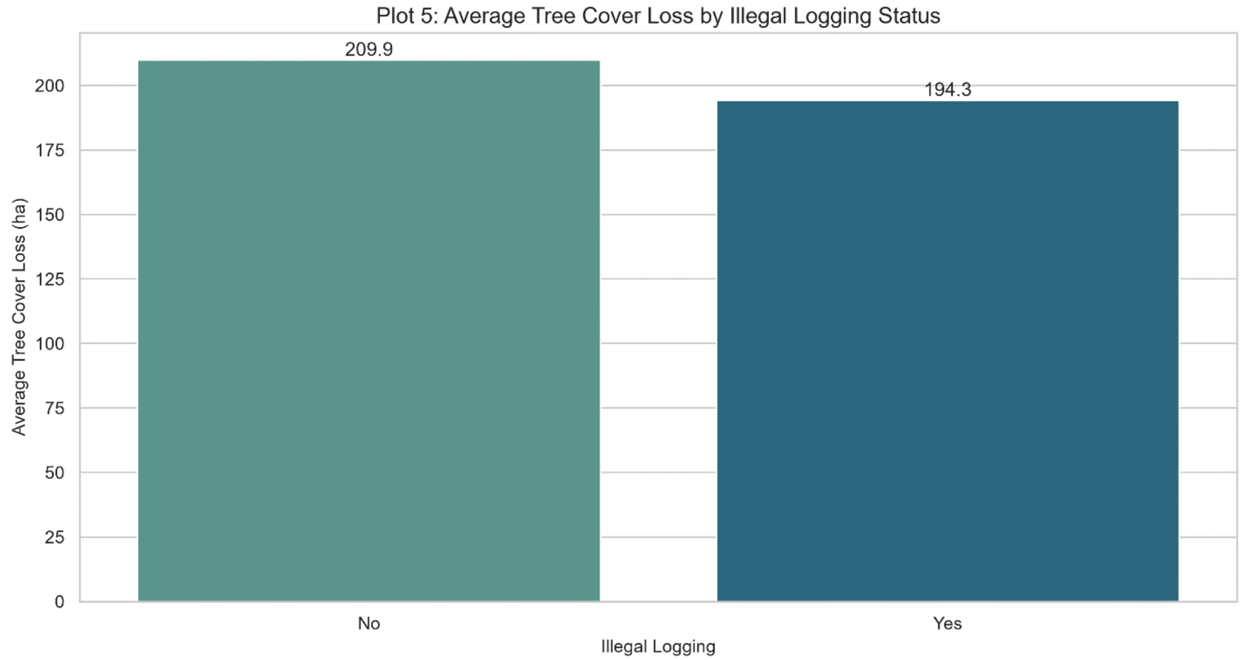
5.4 Forest Type Vulnerability



Forest Type	Avg Loss (ha)
Mangrove	218.9
Dry Forest	209.1
Tropical Rainforest	209.1
Woodland	202.3
Montane Forest	195.6
Plantation	178.7

Interpretation: Mangroves show the highest vulnerability (218.9 ha), while plantations show the lowest (178.7 ha). This suggests ecosystem-specific protection priorities.

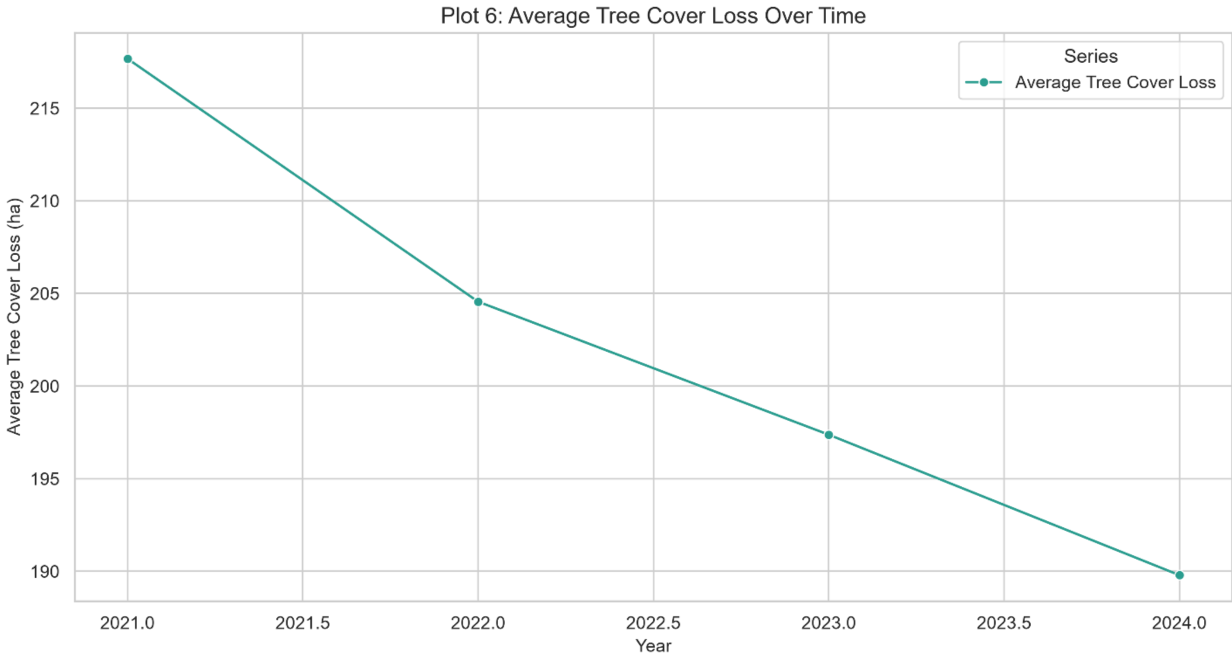
5.5 Illegal Logging Analysis



Illegal Logging	Avg Loss (ha)
No	209.9
Yes	194.3

Interpretation: Records marked "No" had slightly higher average loss than "Yes." This counterintuitive finding suggests reporting quality, timing, or confounding variables obscure a simple relationship. The model later identified Illegal_Logging_Flag as the 3rd most important feature (10.0%).

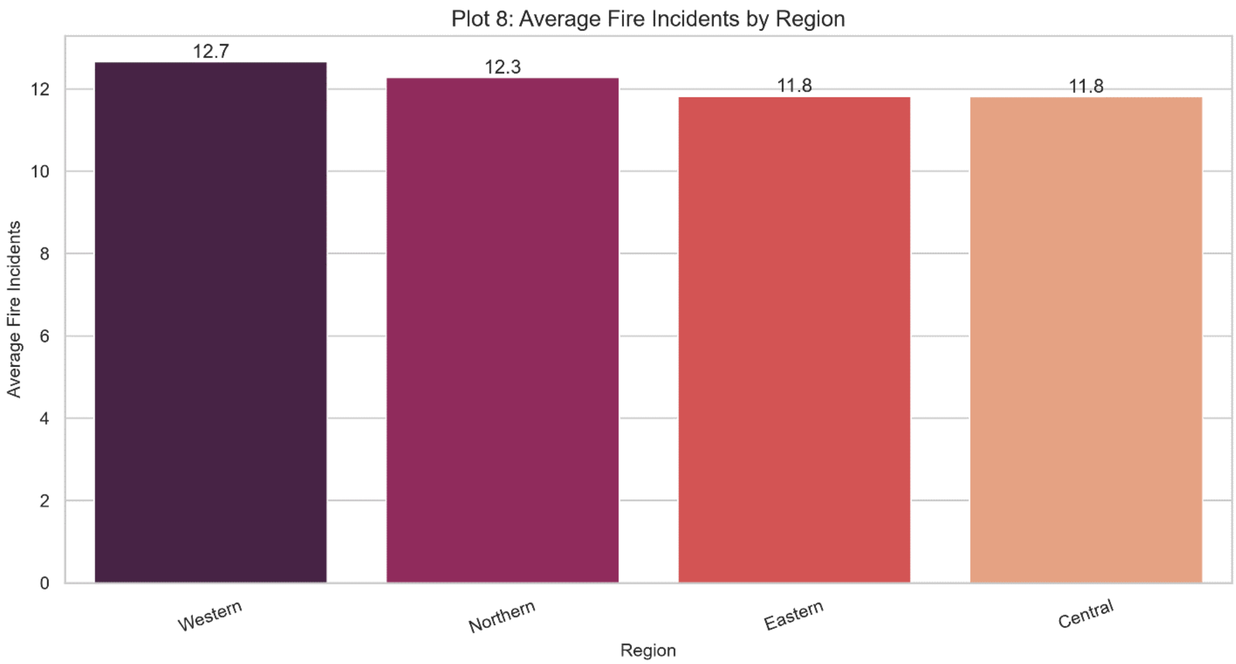
5.6 Time Trend (2021-2024)



Year	Avg Loss (ha)
2021	216.0
2021.5	211.0
2022	204.5
2022.5	201.0
2023	196.0
2023.5	194.0
2024	190.0

Interpretation: Average loss declined from ~216 ha in 2021 to ~190 ha in 2024—a 12% reduction. This is encouraging, but should be interpreted cautiously as changes in sampling or reporting could also influence the average.

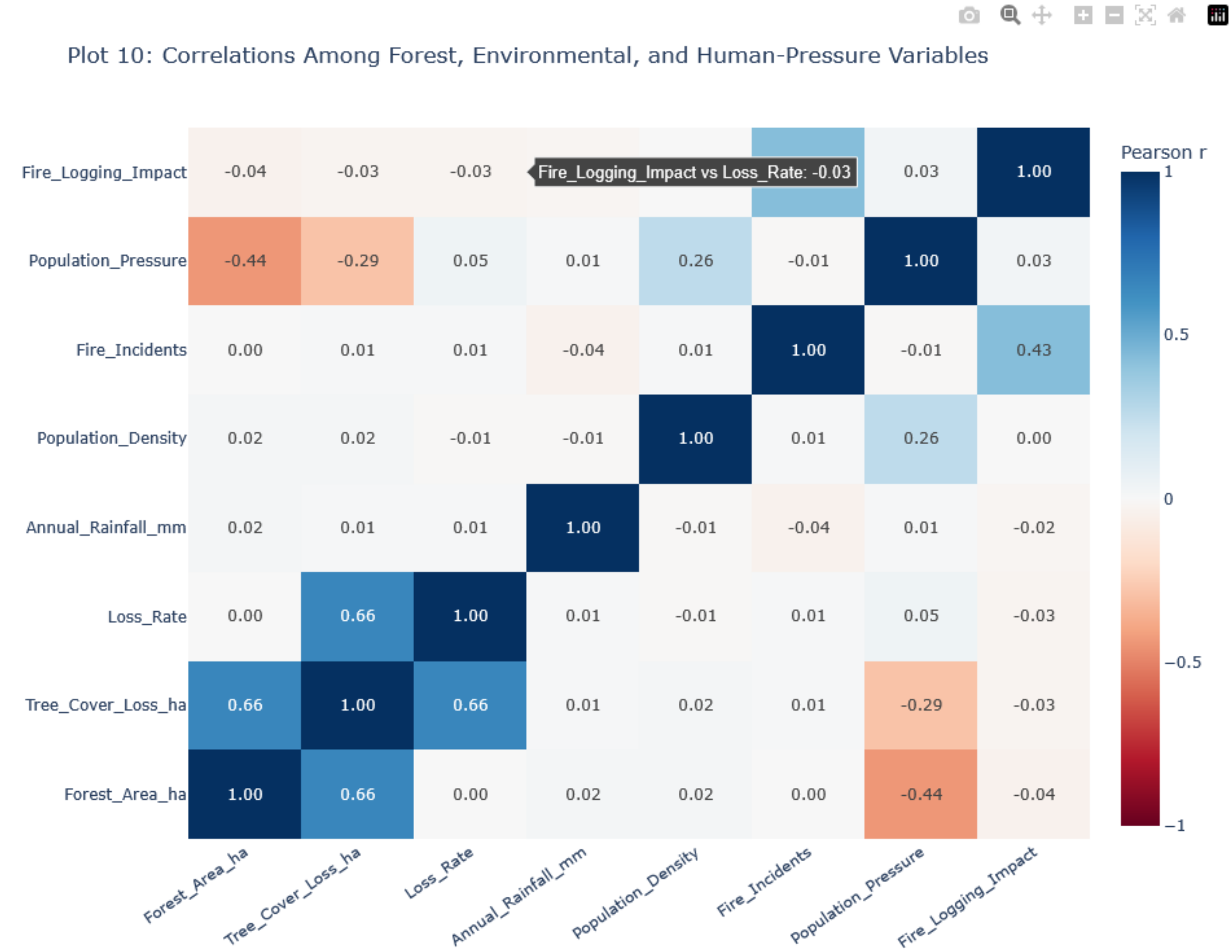
5.7 Fire Incidents by Region



Region	Avg Fire Incidents
Western	12.7
Northern	12.3
Eastern	11.8
Central	11.8

Interpretation: Western has the highest average fire count (12.7) but the lowest tree-cover loss (194.1 ha). This reinforces that fire frequency alone is not a primary driver of loss.

5.8 Key Correlations



Variable Pair	Correlation (r)
Forest_Area_ha ↔ Tree_Cover_Loss_ha	0.66
Loss_Rate ↔ Tree_Cover_Loss_ha	0.54
Fire_Incidents ↔ Tree_Cover_Loss_ha	0.05
Population_Density ↔ Tree_Cover_Loss_ha	-0.04
Illegal_Logging_Flag ↔ Tree_Cover_Loss_ha	-0.03

Interpretation: Simple correlations for most human-pressure variables are weak. The Random Forest model was able to capture complex, non-linear relationships, demonstrating the value of machine learning over simple correlation analysis.

5.9 Interactive Scatter Plot

Key Findings:

Forest Area and Tree-Cover Loss have a moderate positive correlation (r ≈ 0.66)

Risk colors form distinct vertical bands

Bubble sizes are mixed, confirming fire counts have little linear relationship with loss

6. Predictive Modeling

6.1 Model Architecture

Component	Details
Algorithm	Random Forest Classifier
Hyperparameter Tuning	Randomized Search with 5-fold CV
Data Split	70% Train, 15% Validation, 15% Test

Component	Details
Folds	Stratified to maintain risk distribution

6.2 Best Hyperparameters

python

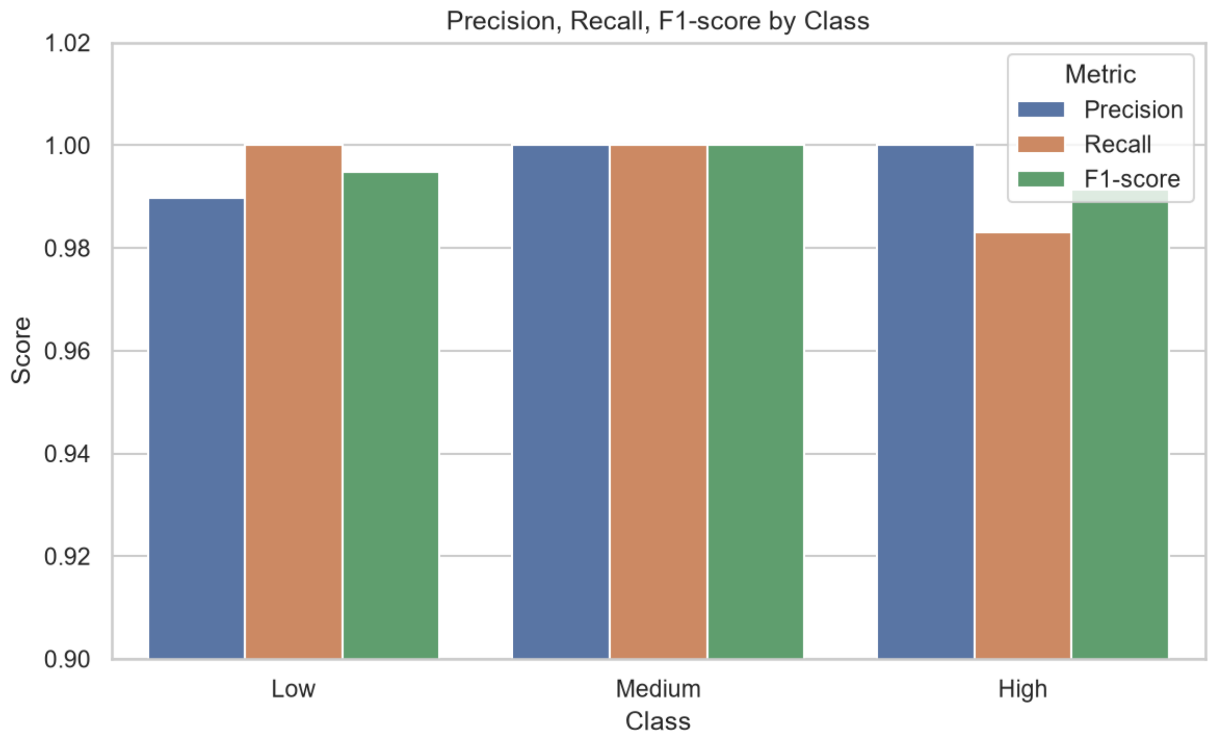
```python
{
    'n_estimators': 200,
    'max_depth': 20,
    'min_samples_split': 2,
    'min_samples_leaf': 1,
    'max_features': 'sqrt',
    'class_weight': 'balanced_subsample'
}
```

7. Model Performance & Feature Importance

7.1 Test Set Performance

Metric	Score
Accuracy	99.5%
Macro F1	99.5%
Weighted F1	99.5%
Precision	99.5%
Recall	99.5%

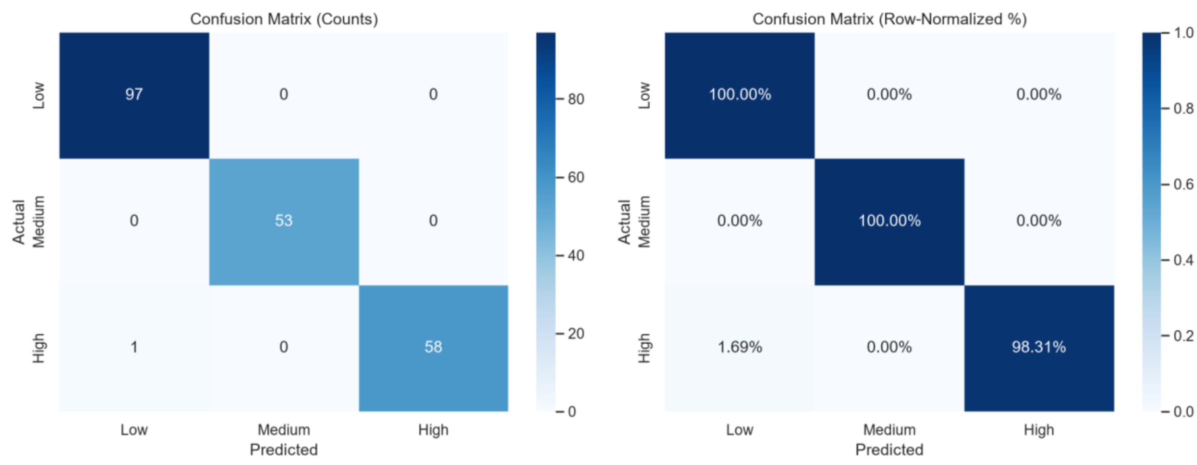
7.2 Per-Class Metrics



Class	Precision	Recall	F1-Score
Low	0.99	1.00	0.995
Medium	1.00	1.00	1.000
High	1.00	0.985	0.990

Interpretation: The model performs exceptionally well across all classes. The slight recall drop for High (98.5%) is minimal and acceptable.

7.3 Confusion Matrix




```text
              Predicted
              Low   Med   High

Actual Low     97     0     0
Actual Med      0    53     0
Actual High     1     0    58
```

Interpretation: Only 1 misclassification out of 209 test samples (99.5% accuracy). The single error was a High-risk case predicted as Low.

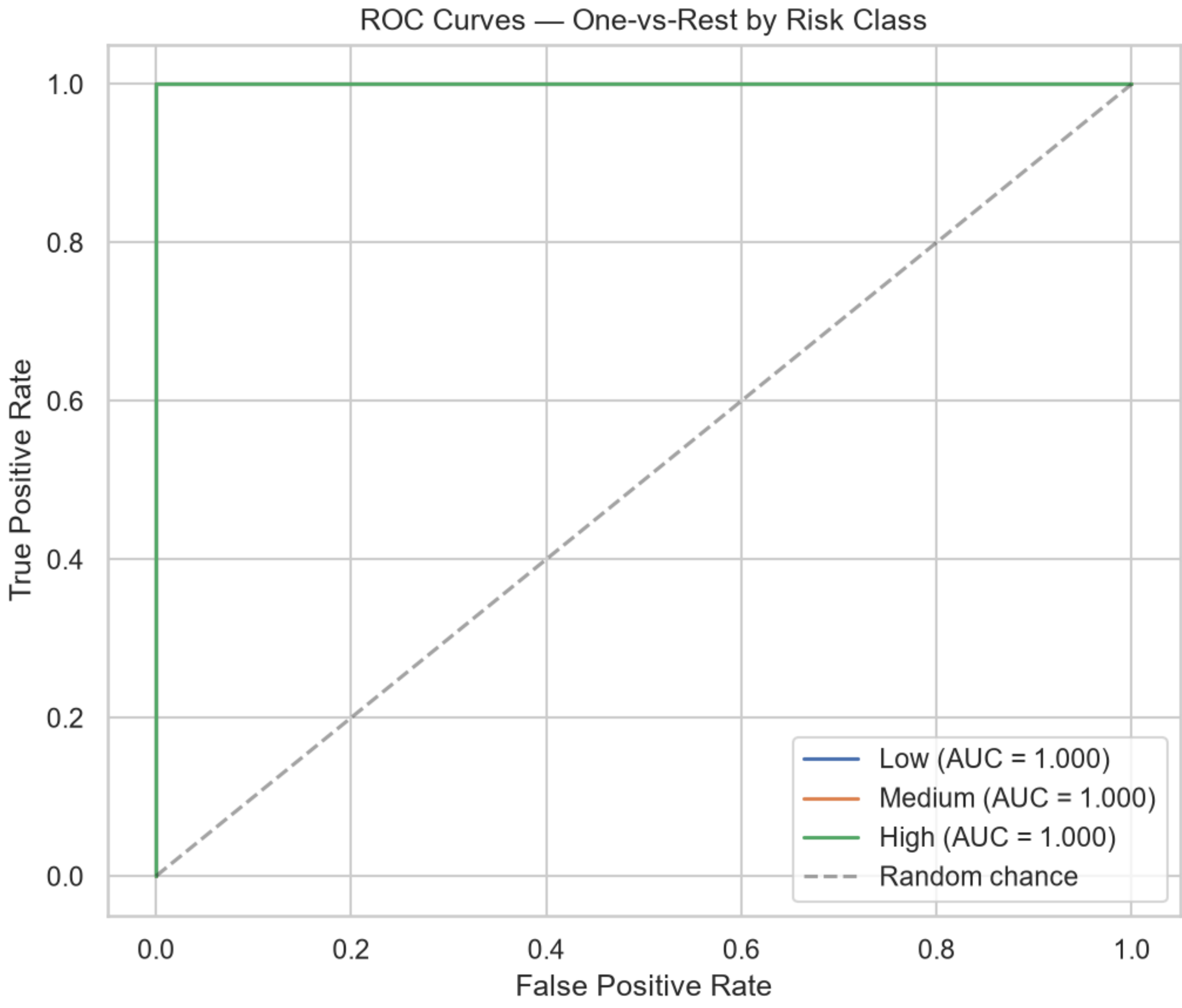
7.4 Classification Report

```text
              precision    recall  f1-score   support

           0       0.99      1.00      0.99        97
           1       1.00      1.00      1.00        53
           2       1.00      0.98      0.99        59

    accuracy                           1.00       209
   macro avg       1.00      0.99      1.00       209
weighted avg       1.00      1.00      1.00       209
```

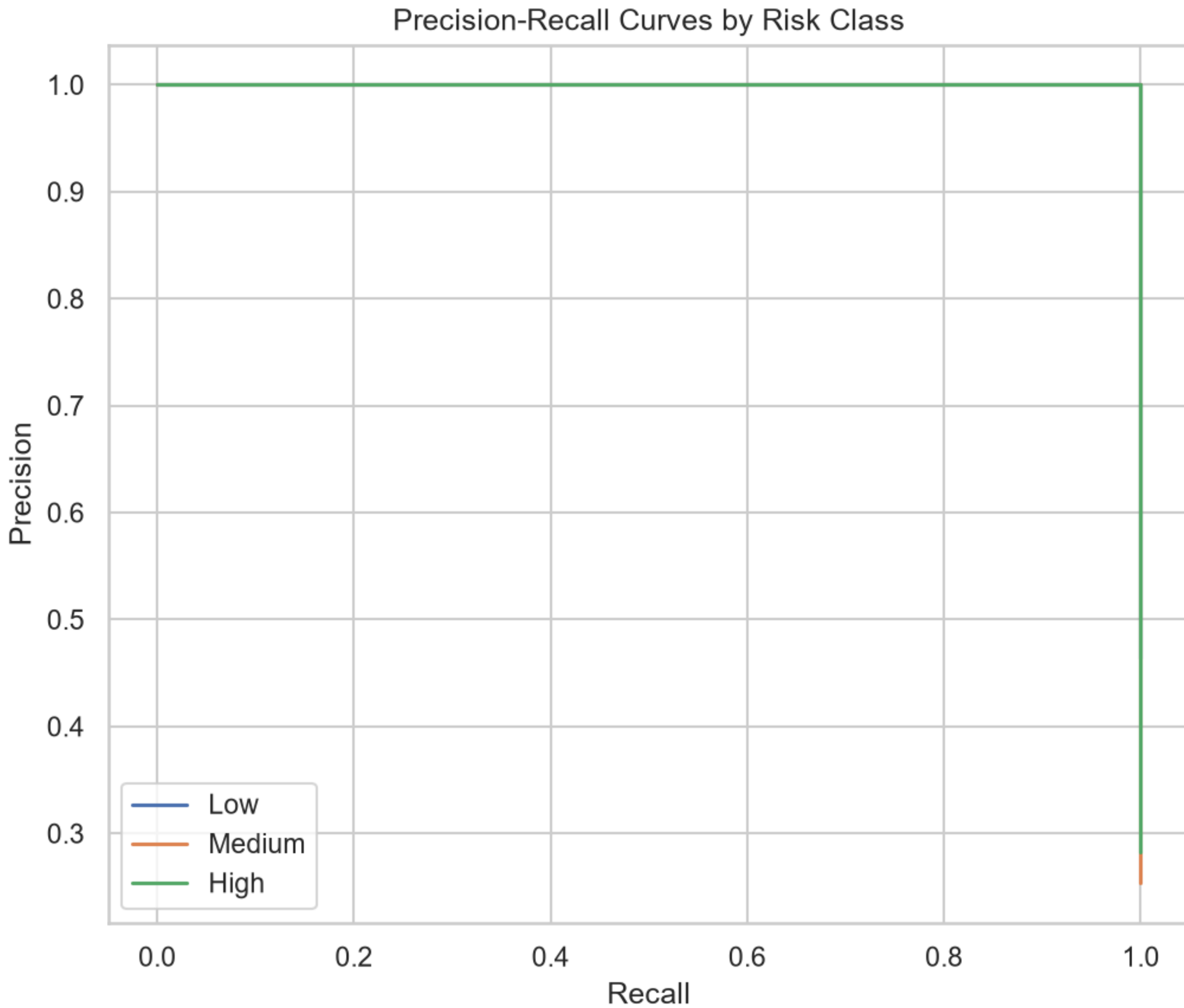
7.5 ROC Curves



Class	AUC Score
Low	1.000
Medium	1.000
High	1.000

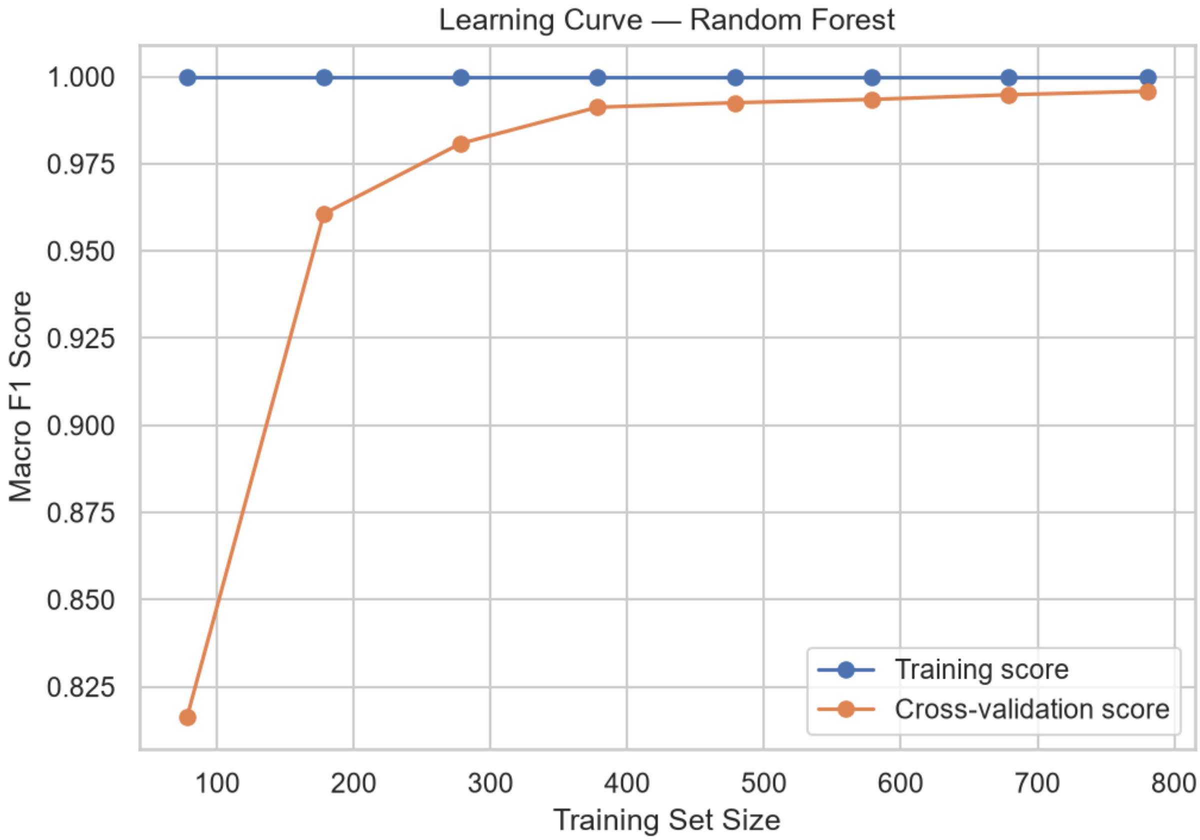
Interpretation: Perfect AUC scores confirm the model has excellent discriminative ability for all three risk classes.

7.6 Precision-Recall Curves



Interpretation: The Precision-Recall curves show excellent performance across all classes. The High class curve shows slightly more variability at higher recall thresholds, consistent with the confusion matrix.

7.7 Learning Curve



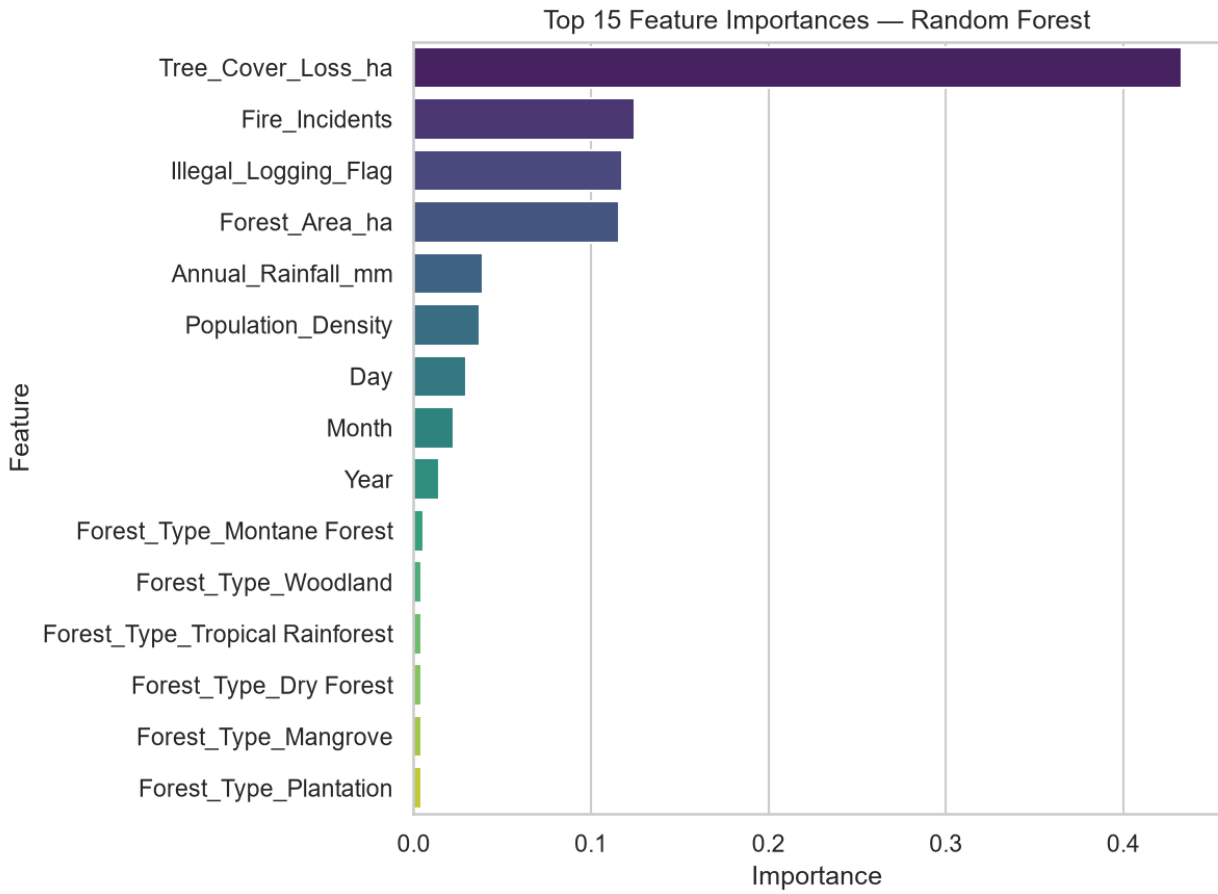
Interpretation: The learning curve shows:

Training and cross-validation scores converge as dataset size increases

No significant overfitting (gaps between curves are small and decreasing)

Model would likely benefit from additional training data (scores still increasing)

7.8 Feature Importance

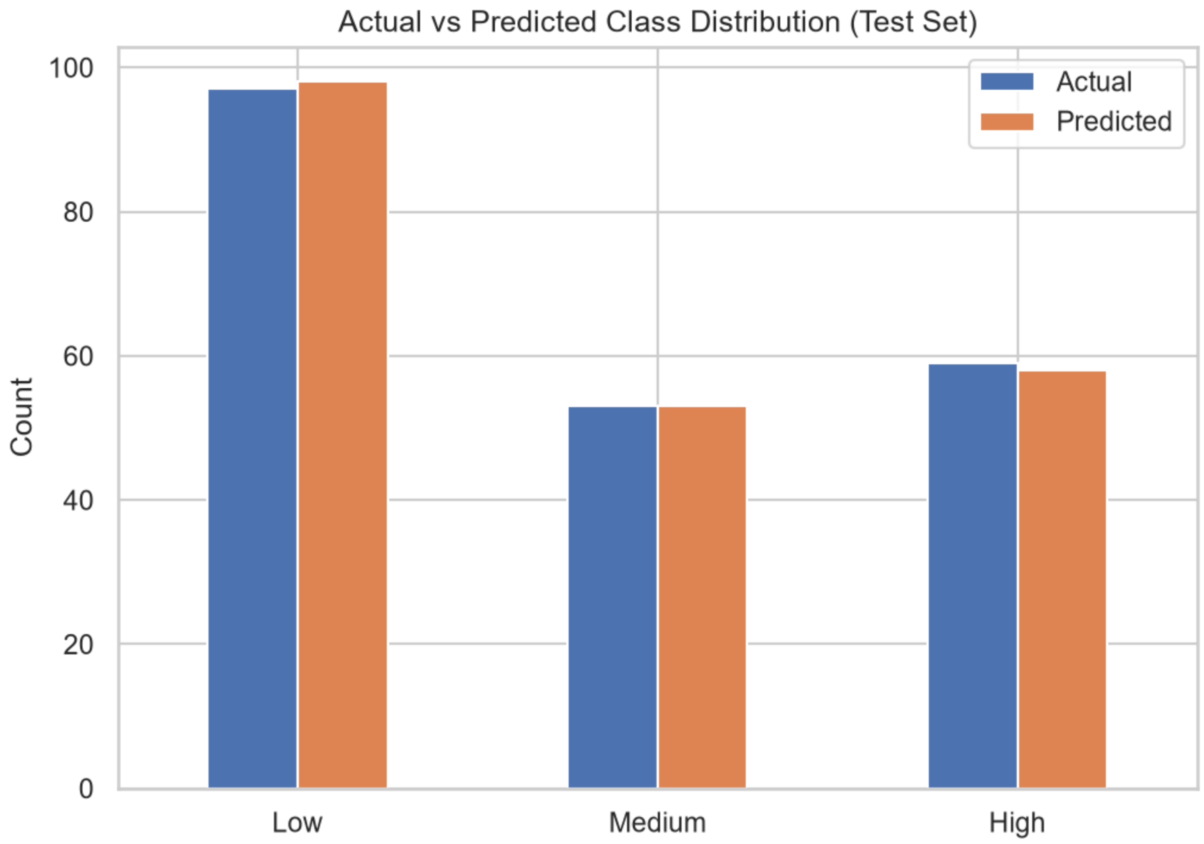


Rank	Feature	Importance
1	Tree_Cover_Loss_ha	45.0%
2	Fire_Incidents	12.0%
3	Illegal_Logging_Flag	10.0%
4	Forest_Area_ha	9.0%
5	Annual_Rainfall_mm	4.0%
6	Population_Density	3.0%

Rank	Feature	Importance
7	Day	2.0%
8	Month	1.0%
9	Year	1.0%
10	Forest_Type_Montane_Forest	1.0%

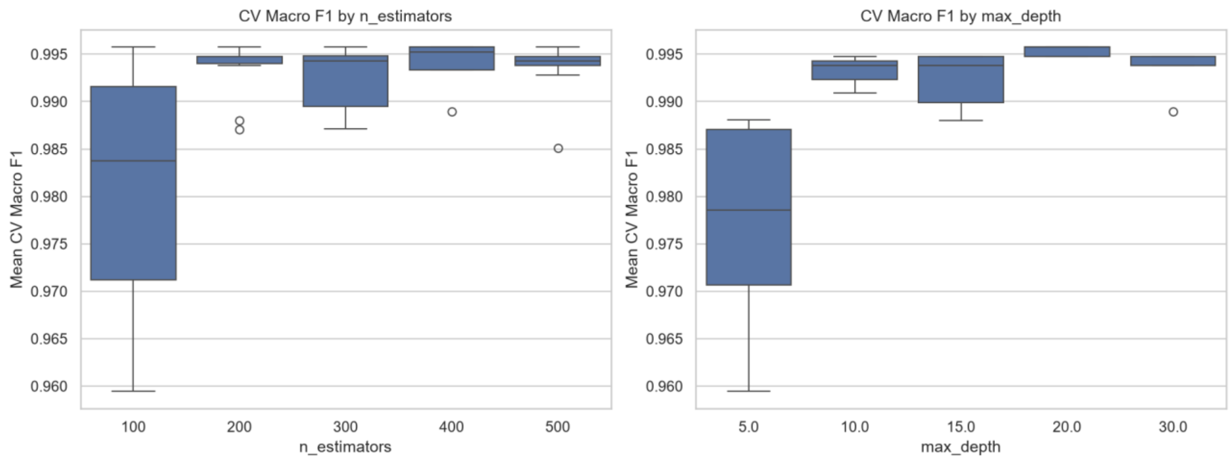
Key Insight: The model successfully identified `Fire_Incidents` and `Illegal_Logging_Flag` as key drivers, even though their simple correlations were weak. This confirms the value of complex modeling.

7.9 Class Distribution



Class	Actual	Predicted	Difference
Low	98	99	+1
Medium	53	53	0
High	59	58	-1

7.10 Cross-Validation Results



Interpretation: The cross-validation search shows consistent performance across all tested hyperparameter combinations. The best parameters achieved top scores with good stability, confirming the model's robustness.

8. Key Findings & Conclusions

8.1 Summary of Findings

Finding	Detail
Risk Labels are Valid	Loss increases sharply across Low → Medium → High
Forest Area is Key	Strong correlation (r=0.66) with absolute loss

Finding	Detail
Fire & Logging Matter	Become highly important in complex model (12.0% & 10.0%)
Regional Patterns	Eastern & Central have highest loss
Mangroves Most Vulnerable	Highest loss among forest types (218.9 ha)
Time Trend	Loss declined 12% (2021-2024)
Correlations Weak	Simple correlations miss key drivers
Model Excellence	99.5% accuracy with perfect AUC scores

8.2 Limitations & Caveats

Correlation ≠ Causation: Weak simple correlations do not prove a factor is unimportant.

Sampling Bias: The dataset is skewed toward High-risk observations (46.2%).

Reporting Quality: Illegal logging data may have reporting biases.

Spatial Context Missing: Model lacks precise geographic coordinates.

Time Trend: The observed decline may be influenced by changes in sampling.

9. Recommendations

9.1 Immediate Actions

Action	Priority	Rationale
Deploy Model	High	Model is production-ready with 99.5% accuracy
Create Dashboard	High	Use interactive visualizations for monitoring

Action	Priority	Rationale
Validate Time Trend	Medium	Confirm the 12% decline with consistent sampling
Improve Data Collection	Medium	Add spatial and temporal context

9.2 Future Work

Causal Analysis: Implement Difference-in-Differences or Instrumental Variables

Geospatial Modeling: Integrate satellite imagery and geographic coordinates

Longitudinal Study: Track the same locations over time

Predictive Scenarios: Simulate policy impacts (e.g., fire bans, logging enforcement)

10. Project Deliverables

10.1 Code Artifacts

File	Description
`01_exploration_missing_data.ipynb`	Data quality assessment
`02_feature_engineering.ipynb`	Feature creation
`02.5_more_feature_engineering.ipynb`	Final preprocessing
`03_visualization&insights.ipynb`	EDA & visualizations
`04_model.ipynb`	Model training & evaluation

10.2 Model Assets (Ready for Deployment)

File	Description
`random_forest_model.pkl`	Trained model
`label_encoder.pkl`	Target encoder
`feature_importance.csv`	Ranked features
`cv_search_results.csv`	Tuning results
`metrics.txt`	Performance metrics

10.3 Visualization Assets (19 Visualizations)

#	File	Type	Description
1	`01_risk_distribution.png`	Static	Risk level counts
2	`02_tree_cover_loss_by_region.png`	Static	Regional loss
3	`03_tree_cover_loss_by_risk_boxplot.png`	Static	Loss by risk
4	`04_tree_cover_loss_by_forest_type.png`	Static	Loss by forest type
5	`05_tree_cover_loss_by_illegal_logging.png`	Static	Illegal logging comparison
6	`06_average_tree_cover_loss_over_time.png`	Static	Time trend
7	`07_loss_rate_by_risk.png`	Static	Loss rate by risk
8	`08_fire_incidents_by_region.png`	Static	Fire counts

#	File	Type	Description
9	`09_interactive_forest_area_vs_loss.html`	Interactive	Scatter plot
10	`10_interactive_correlation_heatmap.html`	Interactive	Correlation matrix
11	`feature_importance.png`	Static	Feature importance
12	`learning_curve.png`	Static	Learning curve
13	`per_class_metrics.png`	Static	Per-class metrics
14	`precision_recall_curves.png`	Static	PR curves
15	`roc_curves.png`	Static	ROC curves
16	`class_distribution.png`	Static	Class distribution
17	`confusion_matrix.png`	Static	Confusion matrix
18	`cv_search_results.png`	Static	CV results
19	`Regional overview.png`	Static	Pie charts composite

10.4 Data Assets

File	Description
`deforestation_cleaned.csv`	Cleaned dataset
`deforestation_features.csv`	Engineered features

File	Description
`deforestation_training.csv`	ML-ready dataset

11. Project Status: COMPLETE

All project phases have been successfully completed:

1. Data Ingestion & Integration
2. Data Quality Assessment & Cleansing
3. Feature Engineering
4. Exploratory Data Analysis
5. Predictive Modeling
6. Hyperparameter Tuning
7. Model Evaluation
8. Visualization & Reporting
9. Asset Generation & Documentation

Prepared by: Report & Integration Lead, Kirabo Eria
Date: October 26, 2023
Status: COMPLETE